Despite his definitive demonstration that Bohm's theory was not impossible, Bell was still concerned about the strangest feature of pilot-wave theory: it was "hideously nonlocal." "Terrible things happened in the Bohm theory," said Bell. "For example, the [paths of] particles were instantaneously changed when anyone moved a magnet anywhere in the universe." Was the nonlocality in Bohm's theory an essential feature of quantum physics? Bell asked this question at the conclusion of his paper demolishing von Neumann's proof, leaving it unanswered as a possible avenue for future work.

Nobody saw Bell's question about nonlocality for a long while; his paper demolishing von Neumann's proof spent two years sitting in an editor's desk due to a series of clerical errors. But Bell couldn't leave the question alone—he wanted to know the answer now. For his next project, he set about finding it. "I knew, of course, that the Einstein-Podolsky-Rosen setup was the critical one [for nonlocality], because it led to distant correlations," Bell recalled later. "So I explicitly set out to see if in some simple Einstein-Podolsky-Rosen situation I could devise a little model that would complete the quantum-mechanical picture and leave everything local."

In his work, Bell used a simplified version of the EPR setup, devised by Bohm in the textbook that he'd written just before developing his pilot-wave interpretation. Bohm's version of the EPR experiment made the whole thing easier for Bell to play around with in his head. Instead of two particles colliding and flying away from each other with entangled momentum, Bohm's version of EPR involved photons with entangled polarization.

Polarization is a property of light—light is an electromagnetic wave, and the polarization is the direction that the wave is doing its waving in. But, for our purposes, all that really matters is that it's directional: polarization is sort of like a little arrow that each photon carries with it that can point in different directions. But it's not quite that simple. For one thing, we can't actually tell what direction a photon's polarization arrow is pointing. All we can do is measure a photon's polarization along one particular axis at a time in a somewhat indirect manner, by shooting it at a polarizer (like a lens in a pair of polarized sunglasses). When a photon hits a polarizer, it either passes through or gets blocked; the closer the photon's polarization is to the